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Backup and Restore

OpenShift – etcd, Namespaces, and OADP

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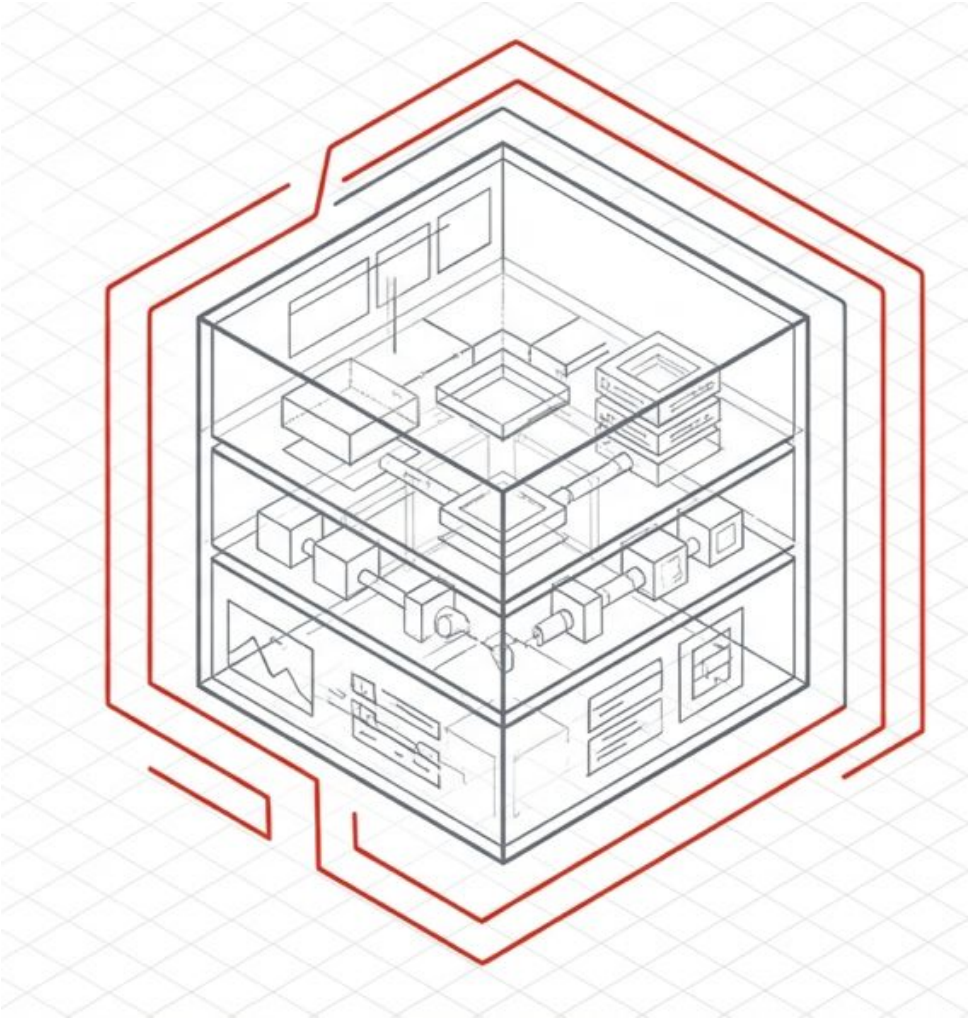
Agenda

- What to protect in OpenShift
- etcd backup and why it matters
- Namespace and workload backup
- OADP – OpenShift API for Data Protection
- Restore scenarios
- Best practices



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What to Protect

Cluster vs applications

- Cluster state: etcd stores Kubernetes and OpenShift API objects
- Application data: PVCs, object storage, external databases
- Configuration: Operators, CRs, secrets (handled carefully)
- Infrastructure: nodes, IaaS – often rebuilt; etcd + data is the focus

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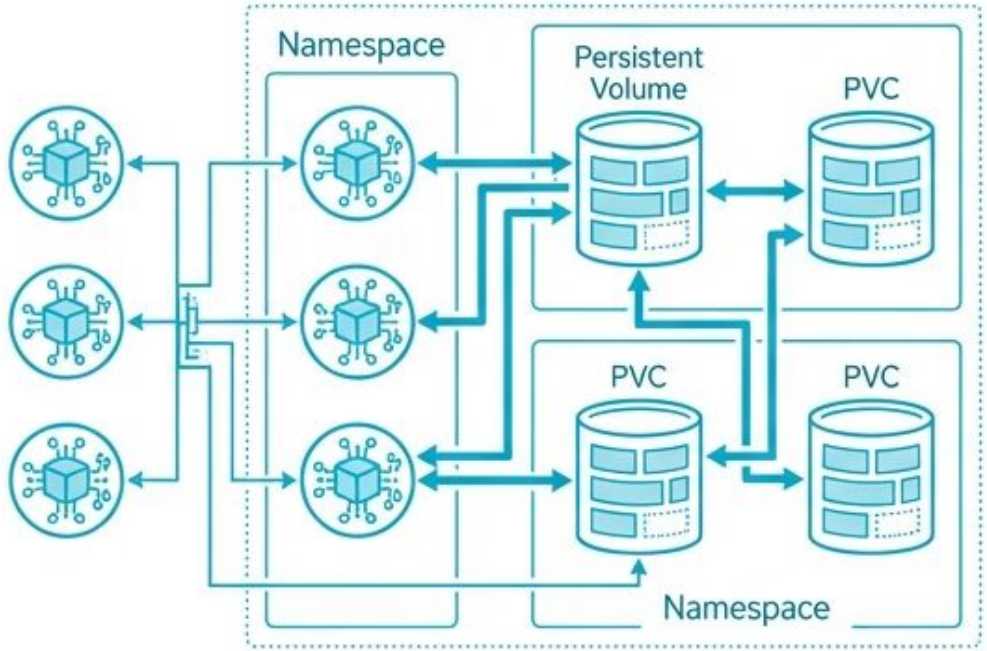
A Cluster is Not Just Compute: The State Divide

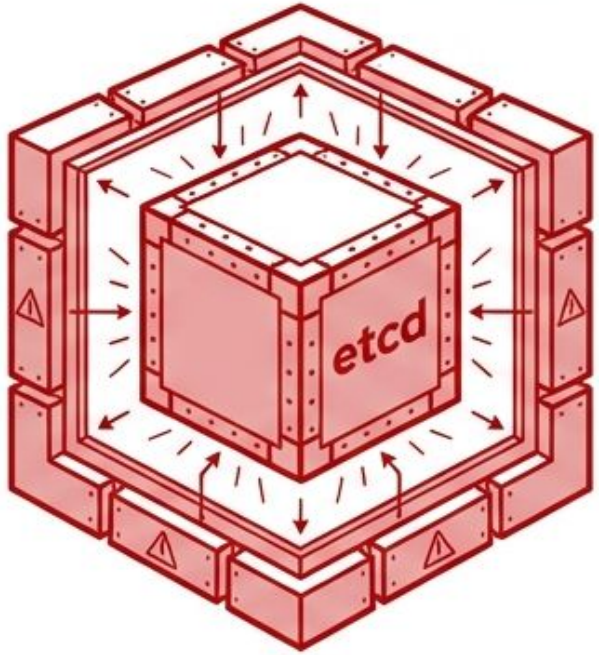
Application Data



The payload of the cluster. Workloads, customer data, and application state. Highly dynamic, cleanly scoped by namespace.

Protected by the OpenShift API for Data Protection (OADP). Routine, non-destructive, and scalable.

Control Plane State



The memory of the cluster itself. The absolute source of truth for all Kubernetes resources, RBAC, and cluster configuration.

Protected by strict etcd snapshots. Restorations are highly destructive, global, and treated as an absolute last resort.

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Disaster Recovery Paradigms: Boundaries of Operation

	OADP (Application)	etcd (Control Plane)
Target	Namespaces, Persistent Volumes (PVs), internal images.	Entire cluster state, core configuration, global resources.
Safety Level	Routine & Safe. Can run hourly without impacting cluster stability.	Destructive. Rewrites history and forces cluster-wide downtime.
Primary Tooling	Velero, Kopia, Container Storage Interface (CSI).	etcd Cluster Operator, manual cluster-restore.sh scripts.
Artifact Storage	S3/Object Storage via BackupStorageLocation.	Secure external vault (NFS/Object) alongside mandatory off-cluster encryption keys.

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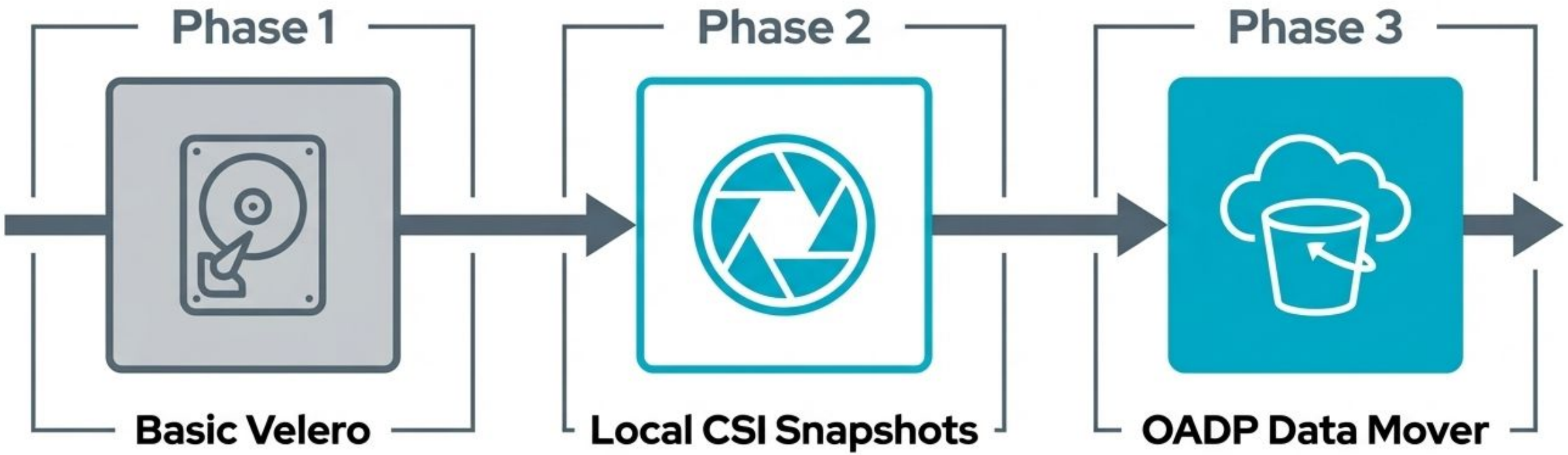
The Evolution of Application Protection



Basic Velero

File-system backups. High overhead, slower recovery, tightly coupled to running pods.

Local CSI Snapshots

Native storage snapshots. Lightning fast, but locally bound. If the storage array dies, the snapshot dies with it. Not true disaster recovery.

OADP Data Mover

The modern paradigm. Takes the speed of a CSI snapshot and physically moves the data out of the storage array into remote object storage. Complete decoupling for true DR.

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The Data Protection Spectrum

	Standard CSI Snapshot	OADP Data Mover
Storage Location	Lives on the primary storage provider.	Remote S3 / Object Store.
Durability Against Infrastructure Loss	⚠ High risk. Artifacts are deleted if the underlying storage or cluster is lost.	✓ Fully durable. Survives complete cluster and storage array annihilation.
Underlying Mechanism	Storage vendor's proprietary snapshot API.	Kopia unified repository engine.
Recovery Scope	Same-cluster or same-storage recovery.	Cross-cluster, cross-region stateful application restoration.

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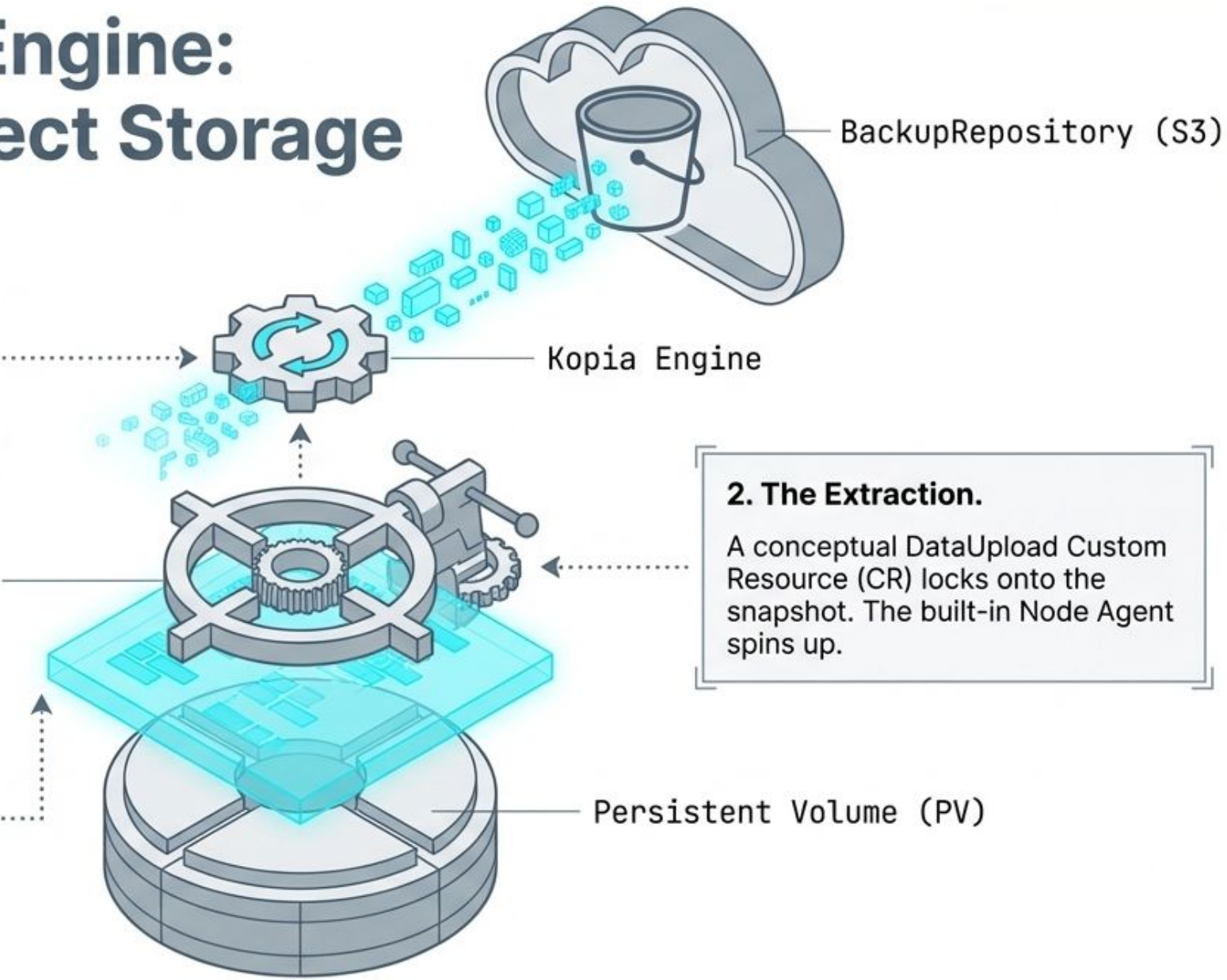
The Data Mover Engine: Extraction to Object Storage



3. The Transport.

The Kopia uploader mechanism reads the snapshot data and streams it into remote object storage. Incremental changes are tracked automatically.

2. The Extraction.

A conceptual DataUpload Custom Resource (CR) locks onto the snapshot. The built-in Node Agent spins up.

1. Locking the State.

An active PV is snapped via CSI. The storage is briefly quiesced, creating a point-in-time local snapshot.

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etcd Backup

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Why etcd Backup

Source of truth

- etcd holds all Kubernetes API resources and cluster configuration
- Disaster recovery often requires a consistent etcd snapshot
- Taken on control plane nodes or via supported automation
- Required before major upgrades and after critical changes
- Restore procedure is platform-specific – follow Red Hat docs

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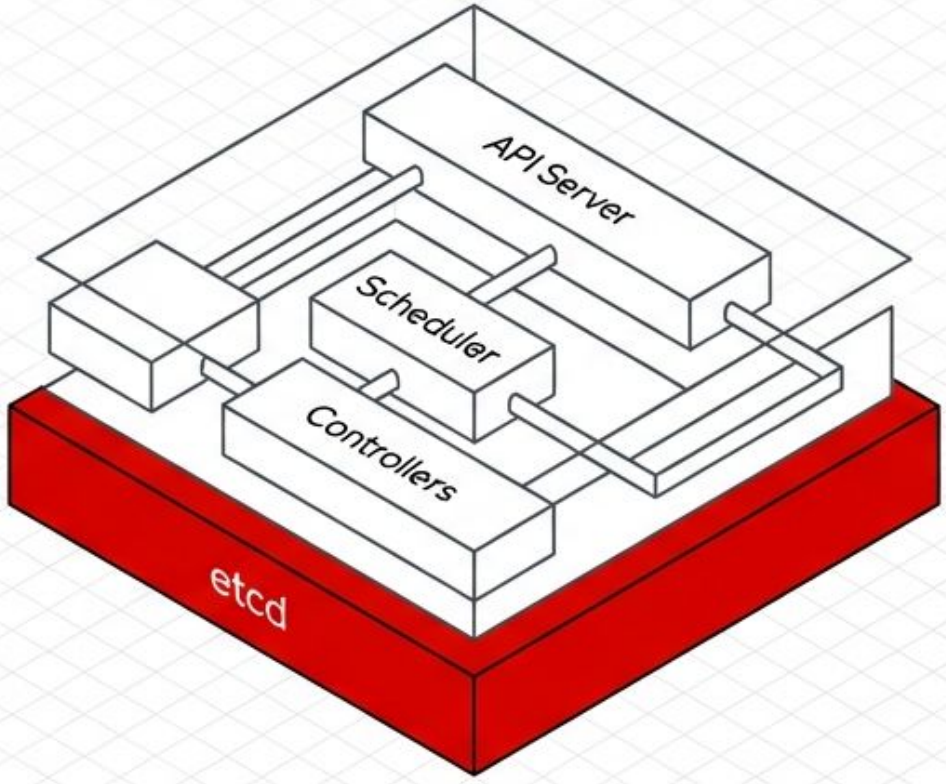
The Blast Radius of State: Protecting the etcd Foundation

The Single Point of Truth

etcd persists the state of all Kubernetes resource objects and configuration data.



Quorum Dependency

Driven by the Raft consensus algorithm, etcd requires a simple majority to function (e.g., 2 of 3, or 3 of 5 nodes).

The Golden Rule

Disaster recovery of the OpenShift cluster is impossible without at least one healthy control plane host or a valid, externally stored etcd backup.

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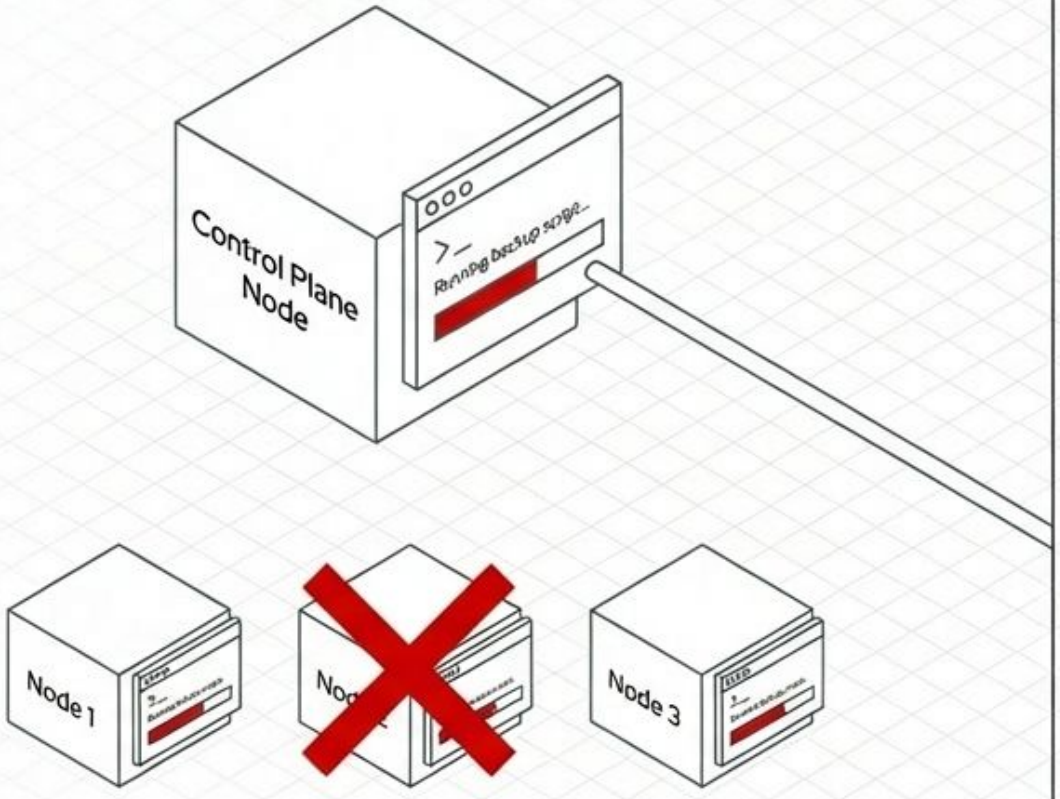
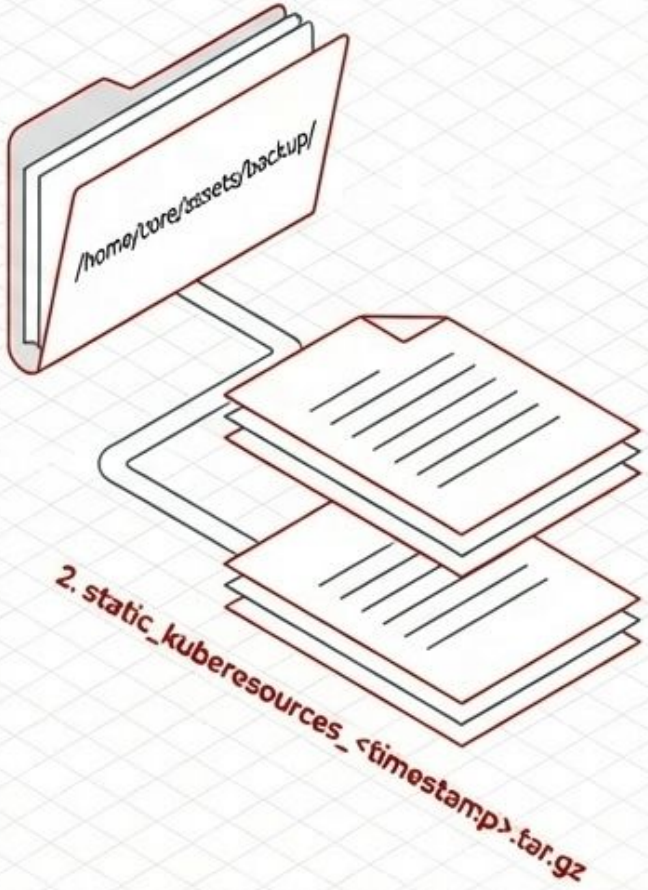
Anatomy of an etcd Backup

The Process



Key Insight: Backups must be generated via a single invocation on a single control plane host to avoid state misalignment.

The Artifacts



1. snapshot_<timestamp>.db
The core database snapshot.

2. static_kuberesources_ _<timestamp>.tar.gz
The static pod resources (and critical encryption keys if etcd encryption is enabled).

Both artifacts are strictly required for a successful cluster restore.

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etcd Backup – Operations

High level

- Schedule recurring etcd snapshots per runbook
- Store snapshots off-cluster, encrypted, with retention policy
- Test restore on non-production before you need it
- Combine with infrastructure backups (VMs, disks) where applicable
- Document who can trigger backup and where artifacts live

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Namespace and Workload Backup

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Namespace Backup

Beyond etcd

- Export manifests: oc get -o yaml (selective, not always complete)
- Velero / OADP: backup API objects and persistent volumes
- CSI snapshots: point-in-time copy of volumes where supported
- EmptyDir and ephemeral data: not in etcd – must be backed up separately
- Secrets: ensure encryption and access control for backup stores

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What Is OADP

Velero-based data protection

- OADP installs Velero and plugins on OpenShift
- Backs up Kubernetes resources and persistent volumes
- Supports object storage as backup target (S3-compatible, etc.)
- Integrates CSI snapshot providers for volume backup
- Red Hat supported operator for application migration and DR

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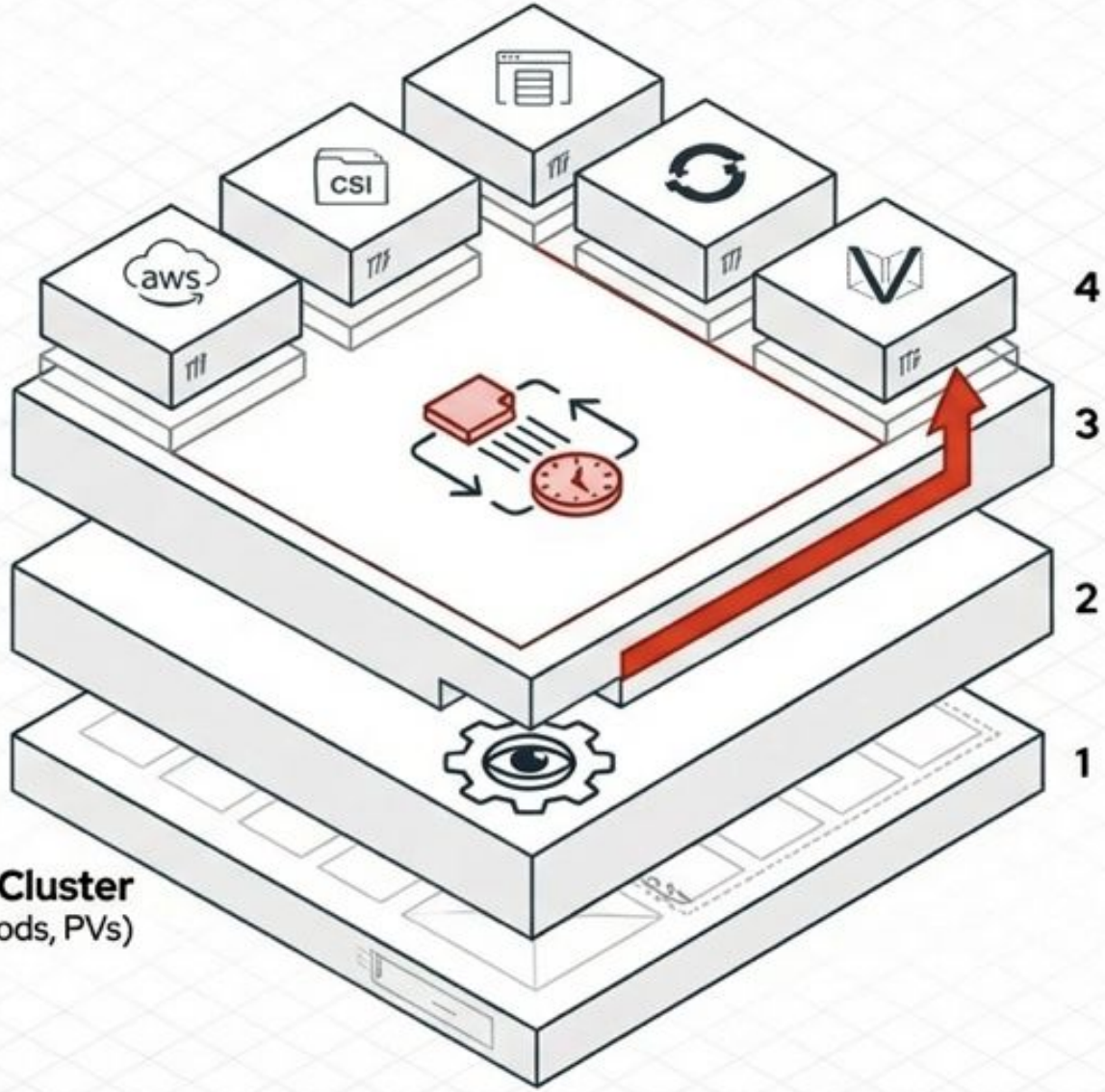
Plugins
(AWS, CSI, OpenShift, OpenShift, KubeVirt)

Velero
(The backup engine, deployed by the Operator)

OADP Operator
(The orchestrator, watching Custom Resources)

OpenShift Cluster
(Namespaces, Pods, PVs)



OADP shifts the paradigm from full-cluster imaging to **namespace-granular workload protection.**

It securely captures Kubernetes resource configurations (state) and internal images, orchestrating **Persistent Volume (PV) capture** via Velero plugins.

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OADP Components

Custom resources

- DataProtectionApplication (DPA): configures Velero and plugins
- BackupStorageLocation: where backups are stored
- VolumeSnapshotLocation: CSI snapshot integration
- Backup: one-time or scheduled job of namespaces/resources
- Schedule: cron-like recurring backups

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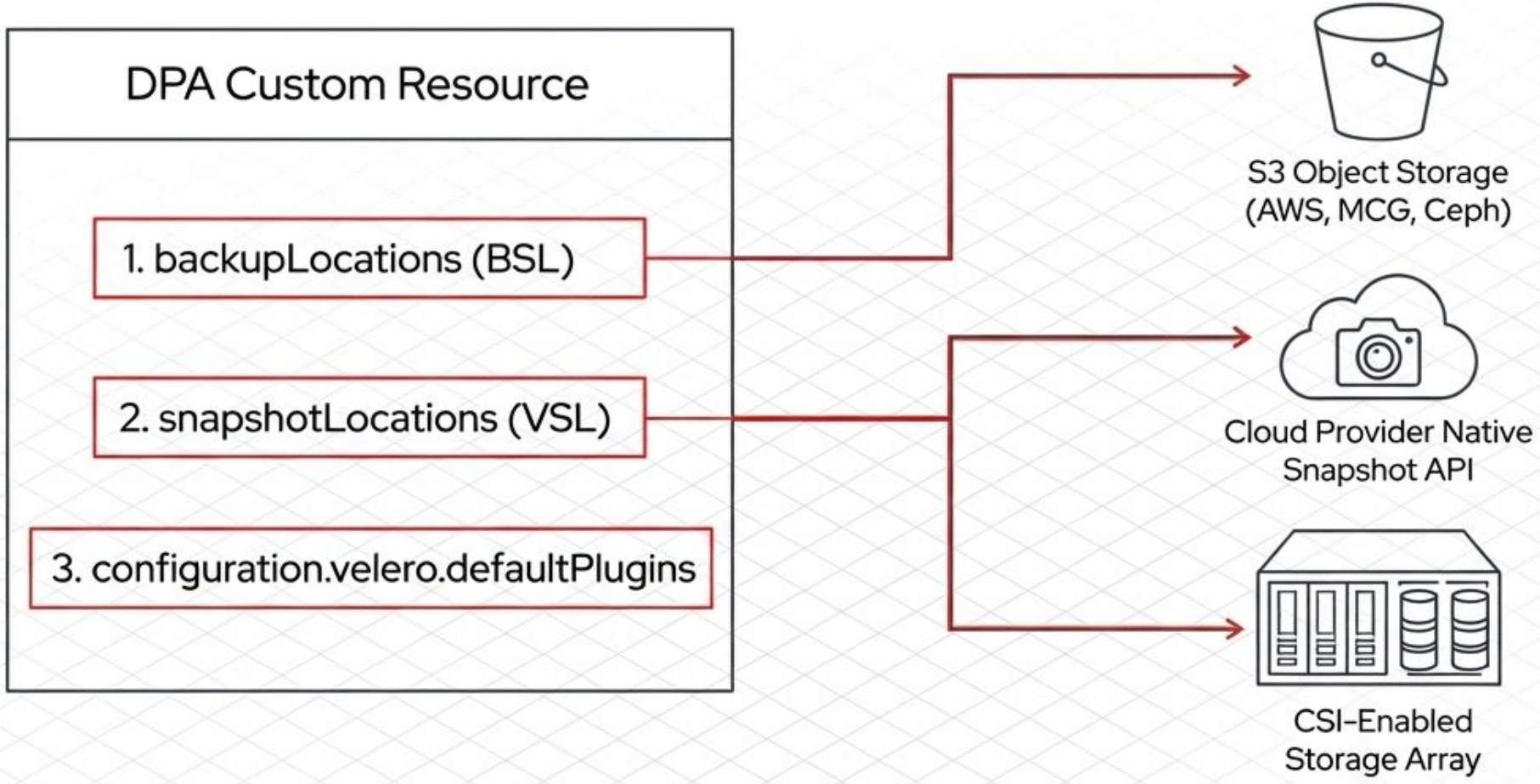
Configuring the Data Protection Application (DPA)



The DPA acts as the central router, mapping internal resource metadata to external S3 BSLs, while directing volume snapshot commands to native cloud or CSI storage interfaces.

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The Volume Protection Matrix

Feature	CSI Snapshots	File System Backup (Kopia/Restic)	Built-in Data Mover (Kopia)
Mechanism	Storage-native snapshot	Pod-level volume scraping	CSI snapshot extracted to Object Store
Filesystem Volume Support	Yes	Yes	Yes
Block Volume Support	Yes	**No**	Yes
Storage Agnostic Migration	No (locked to provider)	Yes	Yes

Insight: While native CSI snapshots are exceptionally fast, they are tethered to the underlying storage provider. True architectural mobility and multi-cloud freedom requires File System Backups or Data Movers.

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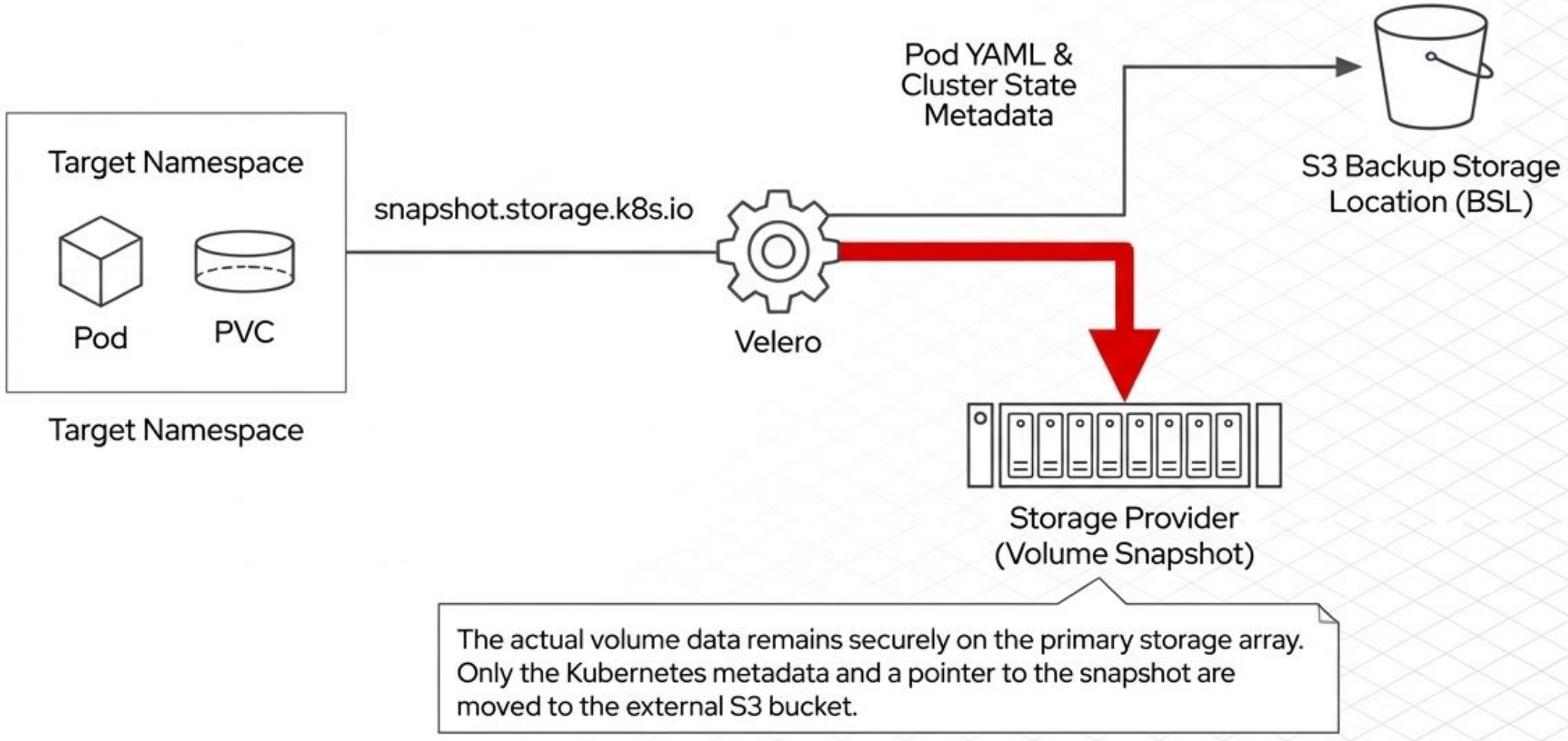
The CSI Snapshot Workflow



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OADP Usage Pattern

Day 2

- 1. Provision object storage credentials and bucket
- 2. Install OADP operator and create DataProtectionApplication
- 3. Create BackupStorageLocation and optional VolumeSnapshotLocation
- 4. Run Backup CR for namespaces or cluster resources
- 5. Use Restore CR to recover to same or another cluster

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Restore

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Restore Scenarios

etcd vs OADP

- Full cluster loss: etcd restore + infrastructure rebuild
- Namespace deletion mistake: OADP restore of namespace backup
- Corrupt app data: restore PVC from snapshot or Velero PV restore
- Cross-cluster: OADP migration patterns with careful UID mapping
- Always validate networking and storage classes on target

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Best Practices

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Backup Best Practices

Reduce risk

- Define RPO/RTO per tier (control plane vs apps)
- Automate etcd backups and OADP schedules; monitor failures
- Encrypt backup data at rest and in transit
- Separate backup admin RBAC from application admin
- Quarterly restore drills with documented results

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Summary

OpenShift backup and restore

- etcd: protect cluster API state; follow supported snapshot procedures
- Namespaces: use OADP/Velero for resources and volumes, not only YAML export
- OADP: operator-based, Backup/Restore CRs, object storage + CSI snapshots
- Test restores; document scenarios for DR and migration
- Read Red Hat docs for your OpenShift version

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Thank You

Questions? See docs.redhat.com – OADP and etcd backup.

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